

Amirhossein Eskorouchi

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SUMMARY

Machine Learning and Optimization Researcher with 4+ years of experience developing novel AI methods and intelligent decision systems across manufacturing, healthcare, and engineering applications. Specializes in computer vision and optimization-guided learning, integrating domain knowledge, physical constraints, and mathematical optimization to develop reliable, interpretable, and decision-relevant AI systems. Proven ability to lead end-to-end R&D, including problem formulation, algorithm design, large-scale experimentation, rigorous validation, and deployment-oriented prototyping, translating research concepts into practical and scalable AI solutions.

PROFESSIONAL EXPERIENCE

Machine Learning & Optimization Researcher

Jan 2024 – Present

Systems Intelligence and Optimization Lab @ Mississippi State University

- Developed UOT-DETR, a PyTorch/DETR computer-vision framework integrating **unbalanced optimal transport** for dense small-object detection and size-distribution estimation; reduced distribution error by up to **30%** and improved small-object recall by **15.4 percentage points** with no inference-time overhead.
- Developed a **3D medical-imaging framework** integrating joint segmentation/classification, anatomically guided learning, and **self-supervised pretraining on 1,500 CT volumes**; achieved **0.928 AUC** and **0.823 Dice** on **150 pathology-confirmed patients**.
- Engineered **Wood-Chip Monitor**, a portable **NVIDIA Jetson/TensorRT edge-AI system** for real-time object detection, size-distribution measurement, moisture classification, quality alerts, and live analytics in biomass manufacturing.
- Built a **graph-based multimodal ML pipeline** for ADHD biomarker discovery using **40K+ polysomnography epochs across 17 physiological channels**; achieved **85% recall** and identified sleep-stage-specific biomarkers.
- Co-developed **PlantLife365**, a multisensor **IoT and ML platform** integrating predictive analytics and an **LLM-based assistant** for real-time agricultural monitoring, alerts, historical visualization, and field decision support.
- Built an **interactive multi-hazard risk-analysis application** integrating damage estimation, annualized risk, **net present value (NPV)** analysis, and geospatial visualization for property and investment decision support.

M.Sc. Researcher

Sep 2021 – Jun 2023

Iran University of Science and Technology, Tehran, Iran

- Developed decision models combining **Dempster–Shafer evidence theory, robust optimization, and scenario analysis** for portfolio, energy, and market decisions under uncertainty; contributed to **5 peer-reviewed studies**.

TECHNICAL EXPERTISE

Machine Learning & AI: Supervised, Unsupervised, and Self-Supervised Learning; Deep Learning; Knowledge-Informed and Constrained Learning; Transfer Learning; Domain Adaptation; Multi-Task Learning; Representation Learning; Graph Learning; Ensemble Learning; Explainable and Interpretable AI; Hyperparameter Optimization

Computer Vision & Imaging: Image Classification, Object Detection, Semantic and Instance Segmentation, 2D/3D Vision, Image-Based Measurement, and Visual Model Interpretability

Optimization & Decision Science: Linear and Nonlinear Programming; Mixed-Integer Programming; Convex and Nonconvex Optimization; Constrained and Robust Optimization; First- and Second-Order Methods; Network Optimization and Dynamic Programming; Bayesian and Multi-Objective Optimization; Optimization for Machine Learning; Optimal Transport and Sinkhorn Methods

Data Science & Analytics: Statistical Learning and Predictive Modeling; Regression Analysis; Time-Series Analysis; Feature Engineering; Feature Selection and Regularization; Dimensionality Reduction; Anomaly Detection; Statistical Inference and Hypothesis Testing; Design of Experiments and ANOVA; Graph and Network Analysis; Community Detection; Business Intelligence and Data Visualization; Database-Driven Analytics

Programming & Technical Stack: Python, R, MATLAB, SQL; PyTorch, TensorFlow, Keras, scikit-learn, MONAI, OpenCV, Hugging Face Transformers, Detectron2, NumPy, Pandas, SciPy; Gurobi, CPLEX, GAMS; TensorRT, NVIDIA Jetson, FastAPI, Streamlit; Power BI, Microsoft Excel, Microsoft Access, SQL Server; Git, GitHub, GitHub Copilot, Docker

AI Engineering & Deployment: End-to-End ML Pipelines; Model Development and Validation; Reproducible Experimentation; Real-Time Inference; Edge AI; Model Acceleration; API-Based AI Applications; Interactive Analytics and Decision-Support Systems

EDUCATION

Ph.D.: Industrial and Systems Engineering, Mississippi State University, MS, USA (GPA: 4.00/4.00) **2024 - Present**
Dissertation: "Evidence-Constrained and Optimization-Guided Learning for Reliable Computer Vision and AI Systems"

M.S.: Industrial Engineering, Iran University of Science and Technology, Tehran, Iran (GPA: 3.52/4.00) **2018 - 2021**
Thesis: "Portfolio Optimization under Uncertainty Using Dempster–Shafer Evidence Theory"

PUBLICATIONS

- **Distributionally Constrained Visual Learning with Unbalanced Optimal Transport**
Decision Support Systems, 2026 (1st Review)
- **Knowledge-Informed Constrained Learning for 3D Medical Imaging**
Medical Image Analysis, 2026 (1st Review)
- **Public Vision Datasets for Wood-Chip Quality Evaluation**
Scientific Data, 2026 (1st Review)
- **Knowledge Transferability in Differentiable Architecture Search for Skin Lesion Diagnosis**
IJSE Transactions on Healthcare Systems Engineering, 2026
- **DA-GMGNet: Domain Adaptation for 3D CT-Based ECE Identification**
IEEE International Symposium on Biomedical Imaging (ISBI), 2026
- **Edge AI for Wood-Chip Quality Evaluation**
IJSE Annual Conference & Expo, 2026
- **Machine Learning-Based Polysomnography Analysis for ADHD Biomarkers**
IEEE International Symposium on Biomedical Imaging (ISBI), 2025
- **DETR-Based Wood-Chip Size Distribution Estimation for Biomass Manufacturing**
Flexible Automation and Intelligent Manufacturing (FAIM), 2025
- **Dempster–Shafer Energy Portfolio Optimization under Deep Uncertainty**
Results in Engineering, 2025
- **Robust Portfolio Optimization Based on Evidence Theory**
Scientia Iranica, 2022

AWARDS, LEADERSHIP & PROFESSIONAL SERVICE

- **1st Place**, IJSE DAIS Student Mobile/Web App Competition, *PlantLife365* 2026
- **Finalist**, IJSE DAIS Student Mobile/Wbe App Competition, *Wood-Chip Monitor* 2026
- **3rd Place**, IJSE DAIS Data Analytics Competition 2025
- **3rd Place**, Campus Vision Challenge, Mississippi State University AI Club 2025
- **Session Chair**, "Advanced AI Solutions in Health and Diagnostics," IJSE Annual Conference & Expo 2025
- **President**, INFORMS Student Chapter, Mississippi State University (*Vice President, 2025–2026*) 2026–Present
- **Mentored 3 graduate and 5 undergraduate researchers** in machine learning and AI, contributing to peer-reviewed journal and conference publications.
- **Peer Reviewer**, *Engineering Applications of Artificial Intelligence, Computers & Operations Research, Journal of Cleaner Production, and Optics & Laser Technology.*